ARCHITECTURE GRAFTING

Exploring DiT Designs via Grafting

Editing Pretrained Diffusion Transformers | 2025

Use a pretrained DiT as architectural scaffolding: distill each replacement operator's activations, then lightly fine-tune the edited graph to study new designs at under 2% of pretraining compute.

DiT · distillation · architecture editing

1 Problem

- Testing a new attention, MLP or depth/width design normally requires full pretraining.
- New operators need a useful initialization, and errors accumulate when many blocks are replaced.
- The question is whether pretrained models can serve as reusable architecture testbeds.

2 Two stages

GRAFTING PROCEDURE

- Stage 1 regresses a new operator onto activations of the pretrained operator using a small dataset.
- L1 regression handles heavy-tailed activation outliers better than L2 in deeper layers.
- Stage 2 performs limited end-to-end fine-tuning to repair accumulated graph-level errors.

3 Design space

WHAT GETS REPLACED

- MHA alternatives: Hyena gated convolutions, sliding-window attention and Mamba-2 linear attention.
- MLP alternatives: expansion ratios 3 or 6 and a convolutional Hyena-X variant.
- Layer choice and replacement ratio matter; interleaved partial grafts are usually most robust.

4 Beyond swaps

RESTRUCTURING

- Self-grafting controls for the procedure by replacing an operator with a randomly initialized copy of itself.
- PixArt-Sigma grafting replaces costly attention on 16,384-token, 2048 x 2048 generation.
- Every pair of sequential DiT blocks can be rewired in parallel, trading depth 28 -> 14 for width.

5 Evidence

QUALITY WITH LITTLE COMPUTE

- ImageNet hybrids reach FID 2.38-2.64 vs 2.27 for DiT-XL/2, using under 2% pretraining compute.
- A 50% Hyena-X PixArt graft gives 1.43x speedup and GenEval 47.78 vs 49.75 on the baseline.
- The 14-layer parallelized model reaches FID 2.77, beating other DiTs of similar depth.

6 Meaning & limits

TAKEAWAY

- Grafting turns expensive architecture research into lightweight editing of an existing model.
- It supports operator replacement, heterogeneous hybrids and graph-level restructuring.
- Results depend on access to a pretrained DiT; synthetic grafting data may add artifacts, and success does not prove train-from-scratch quality.